

Degree Project in Education and Technology
Second Cycle, 30 credits

Artificial Intelligence in Mathematics Education

The Creation and Evaluation of an AI Agent from a Feedback Perspective

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Abstract

The rapid integration of artificial intelligence in education (AIED) has facilitated a paradigm shift from viewing computers as static tools to treating them as symbiotic partners in a collaborative "human-AI alliance". This thesis investigates the application of generative artificial intelligence (genAI) in mathematics education, specifically focusing on the creation and evaluation of an artificial intelligence (AI) agent designed to provide feedback according to the four levels defined by Hattie and Timperley: task, process, self-regulation, and self. Adopting an iterative design process, the study involved the development and testing of multiple agent versions using the Llama 3.3 70B model hosted on a local server at a Swedish upper secondary school. The prompt engineering was guided by the IDEA framework, employing PARTS (Persona, Aim, Recipients, Theme, Structure) and CLEAR (Concise, Logical, Explicit, Adaptive, Restrictive) principles to refine the pedagogical behaviour of the agent. Empirical data were gathered from about 60 conversations with Swedish upper secondary school students and analysed through thematic analysis (TA) to map interactions onto the theoretical feedback levels. The results confirm that generative AI agents are capable of providing feedback across all four levels, though task level feedback remained most frequent due to the core nature of large language models to provide immediate verification. The findings highlighted a successful shift toward process level feedback and the use of "Where to next?" cues to activate student self-regulation. However, a critical "dissolution of self-regulation" was observed in complex word problems, where the agents frequently defaulted to direct instruction, potentially inducing an "expertise reversal effect". While genAI offers powerful potential for immediate cognitive scaffolding, it imposes a significant evaluative burden on learners, who must possess sufficient prior knowledge to detect factual inaccuracies or "hallucinations". The study concludes that the successful integration of AI agents requires a "humans in the loop" framework to mitigate technological dependency and ensure that technology augments, rather than replaces, the student's own cognitive effort.

Keywords: AI, Artificial Intelligence, Artificial Intelligence in Education, Generative AI, Mathematics Education, Feedback, Prompt Engineering

Sammanfattning

Den snabba integreringen av artificiell intelligens (AI) inom utbildning har möjliggjort ett paradigmskifte från att se datorer som statiska verktyg till att behandla dem som symbiotiska partners inom en samverkande "människa-AI-allians". Detta examensarbete undersöker tillämpningen av generativ artificiell intelligens (GenAI) inom matematikundervisning, med specifikt fokus på konstruktion och utvärdering av en AI agent utformad för att ge återkoppling enligt de fyra nivåer som definieras av Hattie och Timperley: uppgift, process, självreglering och person. Genom en iterativ designprocess innefattade studien utveckling och testning av flera versioner av agenten med hjälp av modellen Llama 3.3 70B, huserad på en lokal server på en svensk gymnasieskola. Prompt engineering delen vägleddes av IDEA ramverket och tillämpade principerna PARTS (Persona, Aim, Recipients, Theme, Structure) och CLEAR (Concise, Logical, Explicit, Adaptive, Restrictive) för att förfina agentens pedagogiska beteende. Empiriska data samlades in från ungefär 60 konversationer med svenska gymnasieelever och analyserades genom tematisk analys för att kartlägga interaktioner mot de teoretiska återkopplingsnivåerna. Resultaten bekräftar att genAI agenter är kapabla att ge återkoppling på alla fyra nivåer, även om återkoppling på uppgiftsnivå förblev mest förekommande på grund av stora språkmodellers grundläggande tendens att ge omedelbar verifiering. Tematiska fynd belyste en framgångsrik förskjutning mot återkoppling på processnivå och användningen av ledtrådar av typen "Vart ska jag härnäst?" för att aktivera elevernas självreglering. En kritisk "upplösning av självreglering" observerades dock i komplexa textuppgifter, där agenterna ofta föll tillbaka på direkt instruktion, vilket potentiellt kan framkalla en "expertise reversal effect". Även om genAI erbjuder en kraftfull potential för omedelbar kognitiv stöttning (scaffolding), lägger den en betydande utvärderingsbörda på de lärande, som måste besitta tillräckliga förkunskaper för att upptäcka sakfel eller "hallucinationer". Studien drar slutsatsen att en framgångsrik integrering av AI agenter kräver ett "humans in the loop" ramverk för att mildra tekniskt beroende och säkerställa att teknologin förstärker, snarare än ersätter, elevens egen kognitiva ansträngning.

Nyckelord: AI, Artificiell Intelligens, artificiell intelligens inom utbildning, generativ AI, matematikundervisning, återkoppling, prompt engineering

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List of Acronyms

AFA automatic formative assessment.

AI artificial intelligence.

AIED artificial intelligence in education.

CAI/L computer-aided instructions and learning.

DLE digital learning environment.

EF elaborated feedback.

FIT feedback intervention theory.

FP process level feedback.

FR self-regulation level feedback.

FS self-level feedback.

FT task level feedback.

genAI generative artificial intelligence.

IF interactive feedback.

ITS intelligent tutoring systems.

KCR knowledge of correct response.

KR knowledge of results.

LLM large language model.

TA thematic analysis.

VR virtual reality.

1. Introduction

1.1 Artificial Intelligence in Mathematics Education

Rapid advancements in AI technology has had significant impact on many sectors of society. AI has emerged as a particularly transformative force, reshaping facets of modern life from healthcare and finance to communication and entertainment (Ifenthaler et al., 2024; Salloum et al., 2024; Samala et al., 2025). AI can be defined as the culmination of developments in computing that machines human-like intelligence, characterised by cognitive abilities, learning, adaptability and decision-making. AI aims to craft systems capable of executing tasks that traditionally require human cognition (Chen et al., 2020; Ifenthaler et al., 2024; Samala et al., 2025).

This technological revolution did not begin with making existing tools faster, rather it has ushered in fundamentally new ways of doing things, presenting opportunities and profound challenges for society (Chen et al., 2020; Roll & Wylie, 2016). The education sector, in particular, has become a significant domain for the application and impact of AI. The integration of AIED, is not a recent phenomenon but rather an evolution that has been underway for decades (Salloum et al., 2024). Its origins can be traced back to the introduction of microcomputers and personal computers in the 1970s, which laid the foundation for early forms of computer-aided instructions and learning (CAI/L) (Chen et al., 2020). This was followed by intelligent tutoring systems (ITS) in the 1980s, which were designed to offer adaptive learning experiences tailored to individual student needs. The 2000s saw a surge in machine learning and big data, allowing educational platforms to leverage data analytics for predictive insights into student performance (Salloum et al., 2024). This historical progression shows a trajectory from basic computer-assisted tools to sophisticated, intelligent systems capable of personalising and enhancing the learning process (Chen et al., 2020).

Today, AIED has expanded its reach into nearly every aspect of the educational ecosystem, broadly categorised into three main areas: administration, instruction and learning. In administration, AI has introduced efficiencies by automating tasks such as grading assignments, reviewing student work for plagiarism, and providing feedback, thereby freeing up instructors' time to focus on more direct pedagogical interactions (Chen et al., 2020). For instruction, AI has facilitated the development of powerful pedagogical tools. These include adaptive learning platforms like Knewton, which tailor curricula to individual student needs (Salloum et al., 2024), and intelligent tutoring systems like MATHia and AutoTutor, which provide personalised, one-on-one guidance in subjects such as mathematics (Chen et al., 2020; Salloum et al., 2024). Furthermore, technologies like virtual reality (VR), 3D simulations, and robotics are being used to create immersive and experiential learning opportunities, making abstract concepts more tangible for students (Chen et al., 2020).

The most profound impact of AIED, however, may be on the process of learning itself. Walkington (2025) outlines four key areas where genAI is currently reshaping mathematics education: supporting mathematical problem-solving, providing interactive tutoring and feedback, adapting mathematical tasks to learner needs, and assisting teachers with lesson planning. In this context, genAI can act as a conversational "thought partner" that assists students in confronting complex tasks rather than simply providing automated answers. By leveraging machine learning algorithms, AIED systems can customise and personalise content to align with a student's unique abilities and learning style, which can foster greater engagement and improve knowledge retention. These systems can track a student's progress, identify areas of difficulty, and provide timely, targeted support and feedback (Chen et al., 2020; Salloum et al., 2024). The goal of AI is to move beyond a one-size-fits-all model of education towards a more individualised and adaptive approach that supports every learner effectively (Chen et al., 2020). This vision aligns with a broader shift in educational philosophy, moving away from a narrow focus on domain-level knowledge towards cultivating 21st-century skills such as critical thinking, collaboration, and self-regulated learning (Roll & Wylie, 2016). This has led to a conceptual shift in AIED from simple automation to the augmentation of human capabilities, where the focus is on creating a symbiotic partnership between humans and AI, a concept referred to as "humans in the loop" (Ifenthaler et al., 2024).

The recent emergence of genAI, particularly large language models (LLMs) like ChatGPT, represent a new frontier in this evolution. GenAI models are capable of generating high-quality, human-like content, including text, images, and even software code, from simple written prompts. The release of ChatGPT in late 2022 sparked a widespread discussion and debate within the educational community, with applications ranging from acting as an around-the-clock personal tutor and brainstorming partner to assisting with writing and coding. This new wave of AI tools offers unprecedented opportunities for creating interactive learning experiences but also introduces significant challenges related to academic integrity, information accuracy, cultural bias in training data, and ethical use (Samala et al., 2025; Walkington, 2025).

Within the broad landscape of AIED, the field of mathematics education has been a particularly prominent area of focus (Roll & Wylie, 2016). The structured and rule-based nature of many mathematical concepts makes it a suitable domain for the application of AI-technologies (Roll & Wylie, 2016). As previously mentioned, ITSs such as MATHia have been developed to provide step-by-step guidance on mathematical problems (Chen et al., 2020). Similarly, adaptive learning platforms can create personalised learning paths for students, adjusting the difficulty of problems based on their performance (Salloum et al., 2024). The capabilities of new genAI tools like ChatGPT to explain complex mathematical concepts, generate practice content, and assist with step-by-step solutions further highlight the potential of AI to support learning in mathematics (Samala et al., 2025). Walkington (2025) outlines hat genAI is currently reshaping this field by supporting mathematical problem-solving, providing interactive tutoring, adapting tasks to learner needs, and assisting teachers with lesson planning. In this context, genAI can act as a conventional "thought partner" that assists students in confronting tasks rather than simply providing automated answers. However, in the Swedish context, the reception within mathematics education has been notably cautious. A survey by the Swedish National Agency for Education (Skolverket) reveals that mathematics teachers, in particular, perceive current AI services as less useful compared to colleagues in other subjects (Skolverket, 2024). This scepticism stems partly from the limitations of LLMs in accur-

ately performing mathematical calculations and the pedagogical concern that students might bypass the essential cognitive struggle required for mathematical understanding (Skolverket, 2024). Consequently, while the potential is recognized, the practical integration in Swedish mathematics classrooms remains tentative, with the lowest student usage reported in mathematical problem-solving compared to other tasks (Skolverket, 2024).

Despite these developments, the practical application of AIED still lags behind its anticipated potential (Ifenthaler et al., 2024). There remain significant challenges to its widespread and meaningful implementation, including ethical concerns about data privacy and algorithmic bias, the need for robust technological infrastructure, and pedagogical questions about how to best integrate these tools into existing classroom practices (Ifenthaler et al., 2024; Salloum et al., 2024). Furthermore, there is a recognised need for more reliable empirical research on the impact of AI on educational outcomes (Ifenthaler et al., 2024).

This urgency is reflected in the Swedish educational landscape, where schools grapple with the practicalities of AI integration. Reports indicate a lack of consistent guidelines, with roughly half of upper secondary teachers stating rules for AI use during lessons is lacking in their schools (Skolverket, 2024). To mitigate risks associated with academic integrity and cheating, many Swedish educators have reverted to analogue assessment methods, such as pen-and-paper exams in controlled environments, reducing the reliance on take-home assignments (Skolverket, 2024). The rapid advancement of the technology, especially genAI, means that the educational community is still grappling with its implications, creating an urgent need for research that can inform policy and practice (Samala et al., 2025).

In Sweden, the integration of AI into the educational system is further characterised by a "cautious curiosity" and a strong focus on digital competence. Skolverket (2025) emphasizes that while AI is not yet explicitly mentioned in the general curriculum, it falls under the broader umbrella of digital skills required in schools. To support this transition, Skolverket (2025) published guidelines stressing the importance of a critical approach and the "human in the loop" perspective to ensure safety and learning focus. Despite these efforts, while two-thirds of upper secondary teachers have used AI services, usage remains largely limited to planning and preparation rather than direct classroom instruction (Skolverket, 2024). Teachers also report a significant need for professional development, specifically regarding how to manage and relate to students' use of AI. Marking a significant step towards formalizing this knowledge, a new subject, "Artificial Intelligence", was introduced in Swedish upper secondary schools in the autumn of 2024, aiming to provide students with a deeper understanding of the technology's societal and technical dimensions (Skolverket, 2024, 2025).

1.2 A Didactics Framework for Evaluating AI Generated Feedback

Feedback is widely established as one of the most critical influences on student learning (Black & Wiliam, 1998; Hattie & Timperley, 2007). However, the mere presence of feedback is insufficient to guarantee learning; its effectiveness is highly variable and dependent on the nature of the information provided and how it is received (Kluger & DeNisi, 1996). Pears et al. (2014) emphasize this by demonstrating that students often ignore feedback entirely if it is perceived as a summative judgement rather than a "feed-forward" tool

with immediate relevance for subsequent tasks. This variability highlights the complexity of feedback processes and underscores the need for a theoretical understanding of how feedback influences learning (Hattie & Timperley, 2007).

To address this variability, Hattie and Timperley (2007) proposed a model of feedback that focuses on reducing the discrepancy between a student's current understanding and the desired learning goal. They argue that effective feedback must answer three major questions for the learner: "Where am I going?" (feed up), "How am I going?" (feed back), and "Where to next?" (feed forward). Prompiengchai et al. (2025) link these levels directly to the core pedagogical principles of formative assessment, likening the integration of LLMs to a "GPS-guided learning journey"; the AI provides continuous, adaptive, turn-by-turn guidance, yet the learner remains the driver and the teacher remains the navigator who plans the route. Furthermore, they categorise feedback into four levels: task, process, self-regulation, and self. Feedback directed at the "self" (e.g., praise) is generally the least effective, as it often directs attention away from the task, whereas feedback focused on the process of learning and self-regulation is most effective for deep learning (Hattie & Timperley, 2007).

Wiliam and Leahy (2015) assert that feedback should cause thinking and be "more work for the recipient than the donor". Furthermore, the mere provision of feedback by an AI agent does not guarantee its effectiveness or its acceptance by the learner. Pears et al. (2013, 2014) emphasize that students often perceive feedback as a summative judgment rather than a formative tool, causing them to ignore it entirely if it lacks immediate relevance. They argue that for feedback to be valued, it must adhere to "feed-forward" principles: it must directly enhance student outcomes on subsequent tasks, and the feedback process itself must be highly interactive. If students do not actively engage with the feedback to bridge the gap between their current status and the learning goal, the feedback loop remains incomplete.

In the context of mathematics education and digital tools, Salloum et al. (2024) extend these concepts to "automatic feedback." They distinguish between verification feedback, knowledge of results (KR), which simply indicates whether an answer is correct or incorrect, and elaborated feedback (EF), which provides explanations, hints, or worked examples. While verification feedback is common in digital platforms, Salloum et al. (2024) note that it often fails to support deep conceptual understanding if used in isolation. Instead, EF that guides the learner through the problem-solving process, aligning with Hattie and Timperley's (2007) "process level" feedback, is more likely to foster learning. As Prompiengchai et al. (2025) highlight, current LLM applications frequently fail to distinguish between task, process, and self-regulatory feedback, risking superficial evaluations. Despite the different types of feedback, for the purpose of this study the framework set by Hattie and Timperley (2007) and the feed-forward principles articulated by Pears et al. (2014) will guide the study.

1.2.1 Operationalising Feedback Levels for AI Evaluation

While feedback is intuitively seen as beneficial, the mechanisms governing its effectiveness are complex and multidimensional. In their analysis, Kluger and DeNisi (1996) challenged the assumption that feedback only improves performance, revealing that in over one-third of cases, feedback interventions actually reduced performance. To explain this variance, they proposed the feedback intervention theory (FIT), which posits that the effectiveness of feedback depends primarily on where it directs the recipient's attention. FIT suggests

that attention is regulated hierarchically across three levels:

(1) **Task-learning processes:** Attention is focused on the details of the task and strategies to improve.

(2) **Task-motivation processes:** Attention is focused on the effort required to achieve the goal.

(3) **Meta-task processes:** Attention is focused on the "self" (e.g., self-esteem, ego).

The effectiveness of feedback in digital environments is often determined by its alignment with the learner's cognitive needs. To evaluate AI-generated responses, this study adopts the four levels of feedback defined by Hattie and Timperley (2007): Task level feedback (FT), which focuses on the outcome; Process level feedback (FP), which addresses the underlying strategies; Self-regulation level feedback (FR), which supports metacognitive monitoring; and Self-level feedback (FS), which provides personal evaluations. Barana et al. (2021) argue that moving beyond simple FT is essential for closing the gap between current and desired performance, especially through "interactive feedback" that guides students through the solving process (Barana et al., 2021).

To illustrate how these theoretical levels manifest in a specific mathematical context, Table 1 operationalises them using a common algebraic error.

Table 1

Operationalisation of Hattie and Timperley's Feedback Levels for an Algebraic Error

Level	**Focus**	**Example Feedback for the error:** $3(x-4) \rightarrow 3x-4$
FT	Task	"Your answer is incorrect. The correct solution to the equation is $x = 12$."
FP	Process	"Check your distribution in the first step. You need to multiply the 3 by both the x and the -4."
FR	Self-Regulation	"You have reached a solution. How could you substitute your value back into the original equation to verify if your steps were correct?"
FS	Self	"You are doing a great job; you are clearly very good at algebra. Keep trying!"

Note. Levels of feedback adapted from Hattie and Timperley (2007), applied to a common distributive property error in a linear equation.

In mathematics education, these theoretical distinctions are critical for evaluating current practices. A literature review by Söderström and Palm (2024) found that research predominantly focuses on FT, specifically information regarding the correctness of answers and solution processes. In contrast, feedback targeting self-regulation was found to be rare. This represents a significant gap, as FR is essential for helping students develop the metacognitive skills required to manage the struggle inherent in mathematical problem-solving (Söderström & Palm, 2024).

This gap is particularly relevant when evaluating AI systems within a digital learning environment (DLE). Barana et al. (2021) define automatic formative assessment (AFA) as the use of technology to analyse students' answers and return feedback to reduce the gap between current and desired understanding. They propose a model of interactive feedback (IF), defined as a step-by-step interactive process that guides the learner in resolving a task after one or more autonomous attempts. In this model, the system investigated prerequisites, simpler tasks, or different representation registers (e.g., numeric vs. graphic) to guide the student's dialogue with the problem.

Such interactive strategies are essential for converting external feedback into "inner feedback," a key element of self-regulation (FR). Furthermore, Barana et al. (2021) found that this approach is significantly more effective at engaging learners, particularly those from lower socio-economic backgrounds—in actions aimed at improving their results compared to non, adaptive feedback.

The distinction between task and process feedback is also central to computer-based mathematics instruction. Attali and van der Kleij (2017) differentiate between knowledge of correct response (KCR), which merely verifies an answer (Task level), and EF, which provides explanations or hints (Process level). Their research indicates that while KCR is the most common form of feedback in digital environments due to its ease of implementation, EF is significantly more effective for learning complex tasks. EF supports the learner by explicitly guiding attention to the underlying concepts rather than just the outcome. However, Attali and van der Kleij (2017) and Barana et al. (2021) also note that the timing and design of this feedback are crucial; whereas immediate feedback is generally more useful than delayed feedback for procedural knowledge, it must be carefully structured to prevent students from "gaming the system" or adopting trial-and-error strategies.

1.3 Prompt Engineering

GenAI models operate by predicting text based on learning patterns, meaning the quality of the system's output is directly dependent on the quality of the user's input. This dependency highlights the importance of *prompt engineering*, defined as the iterative process of designing, refining, and optimising text inputs to elicit specific, high-quality responses from AI models. Rather than simple queries, effective prompts function as a form of natural language programming that customises the AIs interaction and outputs (Park & Choo, 2025).

To assist in creating AI agents dedicated to education Park and Choo (2025) proposes the IDEA framework, which can be broken down in four essential pillars:

(1) **Include Essential PARTS:** To minimize ambiguity, prompts must include five core components: Persona (the role the AI assumes), Aim (the specific goal), Recipients (the target audience), Theme (style and tone restrictions), and Structure (the desired format of the output)

(2) **Develop CLEAR Prompts:** The agent's instructions are engineered to be Concise, Logical, Explicit, Adaptive to specific needs, and Restrictive regarding scope and length. This structure ensures the AI follows a coherent flow and reduces the likelihood of irrelevant responses.

(3) **Evaluate and REFINE:** The system acknowledges that prompting is an iterative process. It utilizes strategies such as rephrasing keywords, experimenting with context, and utilizing feedback loops to verify accuracy and alignment with user intent.

(4) **Apply with Accountability:** The prompt engineering process incorporates a layer of ethical oversight, ensuring the agent operates with an awareness of AI limitations, such as potential "hallucinations" or biases, to ensure responsible usage.

With these pillars and an iterative development process in mind an AI agent can be developed, where new versions, and prompt improvements comes through data gathered by user testing. Utilising these pillars as a foundation for the more elaborated prompts.

1.4 The Origin of the Project

This thesis emerged from a collaboration with a teacher from a Swedish upper secondary school, who wanted help studying AIED. Further, the thesis is a collaboration with a research group called K-ULF (Compensatory Teaching for Learning and Research), who strive, in part, to link students' work with practical research (KTH, 2023). The underlying goal has been to investigate aspects of how AI could be used in mathematics education. In discussion, and cooperation, the idea of creating an AI agent that could be used during mathematics education emerged. The idea of how the AI agent was supposed to function and behave came in large from personal experience of creating and testing similar AI agents by the author, in their own mathematical studies. The upper secondary school where the study took place have their own servers which run self-hosted AI models, this allows for complete control over who has access, how data is treated, and creation of local AI agents. This allows conversation data to be recorded locally, and to be used for the thesis work.

1.5 Aim

This thesis contributes to the area of AIED by exploring the application of genAI in mathematics education. To do this an AI agent was created that strives to give feedback according to the four levels of feedback (see Section 1.2.1) presented by Hattie and Timperley (2007). The goal is that the AI agent will respond in a manner similar to how a teacher would respond, and provide the student feedback similar to a teacher. The AI agent can be seen as a tool a teacher could use in their mathematics education, and not a replacement for a physical teacher presence.

1.5.1 Scope

To keep within the limits of this master thesis, the scope of the project had to restricted. The thesis explores the nature of the feedback provided by the AI agent according to the four levels of feedback. The level or contents of the mathematics performed is not of interest to this thesis, although the testing was limited to Swedish upper secondary school students. The AI agent can only use AI models available through the TeachGPT web service.

1.5.2 Research Questions

In order to evaluate if the desired goal of creating a AI agent that can provide feedback according to the four levels of feedback can be achieved, questions that could be answered had to be posed. Firstly, regarding creating an AI agent the following question was posed:

- How can an AI agent be constructed to give feedback according to the four levels of feedback?

This will be answered by how the AI agent is prompted. This leads to the next question posed.

- Based on the four levels of feedback, what kind of feedback does the AI agent provide?

Which leads to being able to understand if the AI agent gives the desired types of feedback. Finally, the last question posed is:

- How can feed back provided by an AI agent support learning in mathematics education?

Which is used to gauge whether the AI agent is suitable as a tool in mathematics education.

2. Methodology

The thesis work was conducted in three phases, with an iterative design process in mind (see Fig. 1). The starting point was in the prompt phase, moving into the test phase, and lastly the analysis phase. After the analysis phase was concluded an iteration loop occurred, going back to the prompt phase.

To create a AI agent a web service called TeachGPT was used. TeachGPT is a service hosted locally in a Swedish upper secondary school, which granted access to the service. TeachGPT gave access to eight different AI models (listed in Table 2) aimed at different tasks. Certain models were not suited as an AI agent, and certain models were not suited for mathematical reasoning.

Table 2
Overview of Available Language Models

Model Name	**Provider**	**Description**
Llama 3.3 70B	Meta Platforms	General purpose. Strong in code and mathematics, supports multiple languages.
QwQ 32B	Alibaba Cloud	Reasoning model. Requires temperature 0.6.
Qwen 2.5 Coder 32B	Alibaba Cloud	Specialized model optimized for coding tasks.
DeepSeek R1 Distill 8B	DeepSeek	Reasoning model. Does not support system prompts (unsuitable for agents).
Llama 3.2 1B	Meta Platforms	Lightweight model intended for testing.
Llama 3.2 3B	Meta Platforms	Small model intended for testing.
Pixtral 12B	Mistral AI	Multimodal (image) model. Under development (API use).
Qwen 2.5 0.5B	Alibaba Cloud	Lightweight model intended for testing.

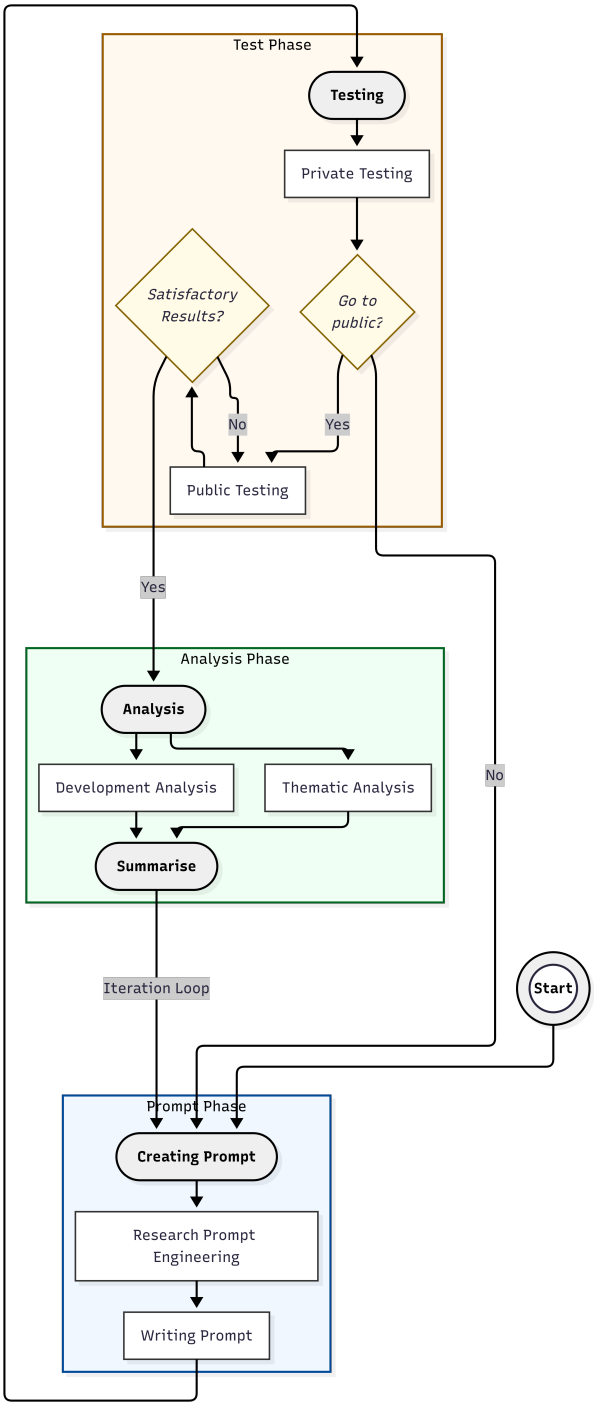
Note. Available AI models on TeachGPT with their respective description and provider.

2.1 Prompt Phase

The prompt phase started of with researching how to create a prompt to achieve the desired goal of creating an AI agent that provides feedback according to the four levels of feedback (Hattie & Timperley, 2007). The research section of the phase took slightly different appearances dependent on when in the cycle it occurred. The writing part of the prompt phase sought to write the prompt that created the AI agent. The writing was done in two steps, first the prompt was written down in a file that was stored in a private

Figure 1

Iterative workflow cycle



Note. Workflow split into three phases. Prompt phase, Test phase and Analysis phase, with starting point in Prompt phase.

GitHub repository, secondly the prompt was written on the TeachGPT web service thus creating the AI agent, any additional settings were also specified at this point.

2.2 Test Phase

The test phase was constructed in two parts, *private testing*, and *public testing*. Private testing was conducted by testing the created AI agent without any outside influence. The author alone tested with material they found to be suitable. After the private testing was concluded a decision had to be made, either to continue to public testing or go back to the prompt phase (as seen by the 'Go to public?' decision point in Fig. 1).

Public testing tested the AI agent on upper secondary school students during their mathematics class. Each student that participated was granted access to the service TeachGPT and the created AI agent that was to be tested. The public testing was then followed by a second decision point, 'Satisfactory Results?' node in Fig. 1, made to make sure the results was suitable for analysis or if more testing needed to be done. Unsuitable results were results of the kind where students did not use the AI agent in a mathematical setting, where they deliberately tried breaking the AI agent, or simply prompted the AI agent with non-mathematical related content.

2.3 Analysis Phase

The two parts of the analysis phase were conducted in parallel with each other and did not directly relate to each other. The *Development Analysis* was an 'analysis' through an engineering lens, it consisted of observing how the AI agent generated responses, how responses were formatted, language usage, and similar aspects that could be used to improve the next iteration of the AI agent.

The *Thematic Analysis* was the larger part of the analysis phase, it sought to analyse the student-AI conversations in how they provided feedback from the perspective of the four levels of feedback presented by Hattie and Timperley (2007).

2.4 Ethical Considerations

An ethical discussion with collaborating teachers from the upper secondary school, hosting TeachGPT, was conducted, with the main discussion points of; how to best measure the AI agents performance, the recording of conversations and its effect on the students.

One idea was to have one group of students work with the AI agent throughout a semester and see how the students performed, all while comparing to a second group of students that would not be using the AI agent. It was deemed that it could lead to one group of students getting an unfair advantage, or that the students felt like they were being treated differently, or unfairly, compared to their fellow students. The decision that participating in the study by using the AI agent would be completely voluntary was made, that way anyone wanting to try the AI agent would be granted access. It was also decided that the best metric to analyse was to perform a thematic analysis on the conversations generated between student and agent, in doing so the feedback given could be analysed.

In order to have conversation data to study all interactions with the AI agent had to be recorded and stored, TeachGPT had a feature to do just this, this also meant that all recorded conversations was stored locally on the servers of the upper secondary school hosting teachGPT. It was considered important that all students using the AI agent was informed that whatever they wrote would be recorded. This was achieved by telling the students that their conversation would be recorded, before they were given access to the AI agent, as well as a message appearing when they start a conversation.

Another consideration was whether students would feel observed, knowing that their conversations were being recorded. The underlying concern being that it could lead students to behave in a way they would normally not when given a tool such as the AI agent. Irrespective of whether or not this were the case, nothing could have been done about it.

2.4.1 Ensuring Anonymity

All students that participated in the study either had or were allocated an account to access the TeachGPT service. Students that had an existing account had their name attached to the account, when a conversation was recorded the author of the AI agent could see the name of each student. Those students who did not previously have an account were granted a demonstration account that had no name associated with it. All conversations that were used, and/or transcribed, were inspected to ensure no personal data were recorded. For transcribed conversations the notation *User* was used for a student user, and *AI* for the AI agent.

2.5 Cycle One

The cycle started at the Starting Point, indicated in Fig. 1, with creating the first prompt. It was decided early that a presentation was going to be held at an upper secondary school, to all year 10 students. The presentation would present the thesis project to the students, inviting them to participate in the project, and to use the AI agent. This was coordinated by teachers at the school to gather all year 10 students at one time, about 120 students. Before the presentation could take place a first iteration of the AI agent had to be created.

2.5.1 Prompt Phase

The research section on the prompt phase consisted of selecting an AI model out of the available options (see Table 2). The model of choice was Llama 3.3 70B from Meta, due to it being a LLM with general knowledge, supporting multiple languages, and being strong in mathematics. Next, was to understand how the AI agent creation was done. TeachGPT has an AI agent creation form (see Appendix A) that handles the creation of an AI agent. The form asked for a name of the agent, which AI model to use, setting a temperature, a system prompt, an introductory question, and a toggle to record user conversations.

The second section of the prompt phase (Fig. 1), *Writing Prompt*, the system prompt had to be written. To test out how the creation worked and understanding the capabilities a simple prompt was made (Appendix B). The idea of the prompt was that it should have captured the essence of the desired behaviour, or persona and aim as described by Park

and Choo (2025), being an assistant that helps the student along the way, providing feedback and not providing direct answers. It was also unknown what kind of feedback the AI agent would provide without being directed to certain types of feedback. Swedish was picked as language for the prompt to try and enforce the AI agent to speak Swedish, considering the users would be Swedish students.

2.5.2 Test Phase

With a first version of the AI agent, private testing could start (Fig. 1). To conduct the first round of private testing, suitable material had to be found. With the targeted students being year 10 students of upper secondary school doing the mathematics one course, exercises of that level had to be found. The website *matteboken* was selected for its wide range of publicly available material in all levels of mathematics in upper secondary school and lower. The exercises chosen were randomly selected from different categories, such as negative numbers, arithmetic, algebra. Results from the private test was deemed to not be satisfactory, hence the 'Go to Public?' decision point (Fig. 1) was answered *no*, and a new prompt had to be created.

2.5.3 Second Prompt Phase

The second prompt expanded on the first by adding more details to how it should behave, mainly adding what Park and Choo (2025) describes as theme, that the AI should provide guidance along the was and not give out answers (Appendix C). With the intended goal of reducing unnecessary FT level feedback.

2.5.4 Second Test Phase

Using the same material as the previous test in the cycle, the private testing was conducted successfully. The first public test could be held. The plan for the public test was that all year 10 students at an upper secondary school would be granted access to the AI agent at the presentation, and after, this would allow them to continuously use the AI agent throughout their studies.

A presentation was created that presented the master thesis project, why it was being done, who the author was, and what the goal of the project were. The presentation was designed to be interactive with five questions spread throughout the slides. All year 10 students, approximately 120, were invited and gathered to partake in the presentation. The presentation was held and the AI agent was shared on site with a link in the presentation, and a QR code that the students could scan. The students were told that participation in the study was completely voluntarily, however encouraged to use the AI agent in their mathematics education. After the presentation a link to the AI agent was sent to a teacher at the school who then forwarded the link to all the students.

After the presentation was held the thought was that the students would start using the AI agent. The students were sent a link to the AI agent on email as well as the mathematics teachers at the school, the email was sent by a teacher at the school.

A week after the presentation had been held no user data had been gathered outside a few students who had opened the AI agent during the presentation. A teacher was contacted at the school to inform that no data was being gathered, who then proceeded to encourage the students to start using the AI agent. After another two weeks had

elapsed and no data had been gathered it became very clear that a different approach was needed, this meant that fewer cycles than intended with bigger improvements between cycles would be needed.

2.5.5 Third Prompt and Test Phase

To adapt to the new situation a small alteration was made to the system prompt. The new version 3 prompt (see Section 3.1.1) was created with the goal of doing what Park and Choo (2025) describes as making the prompt more clear, reducing redundant words. This new version passed the private testing and could be pushed to public testing. One additional change was made in the creation of the third version of the AI agent, a starting 'question' was added. The added 'question' was the phrase 'Hej!', this caused the AI agent to type out a response to 'hej' which resulted in a friendly greeting asking how it could help the user.

An offer was made by a teacher at a different upper secondary school, who offered to have one class where the AI agent could be tested. During the class the students were presented with the thesis work, were informed on how they could participate, and that the conversations they had with the AI agent would be recorded. The students where of year 12 in upper secondary school, their mathematics teacher had prepared material suitable for the class (Appendix D). The students were given freedom in how they solved the problems, which was intentionally at a C-A level problems, they could cooperate, work alone, however when they got stuck they were encouraged to first consult the AI agents before consulting the teachers present.

A total of 28 conversations were recorded which were deemed to have 'Satisfactory Results' (Fig. 1), and the analysis phase could begin.

2.5.6 Analysis Phase

In preparation of the data analysis, all conversations were read through, and a selection was made for the analysis. Conversations that were largely of the type where the student used the AI agent as a search engine, or a calculator, or similar use cases that is not the AI agents intended use, were ignored. One conversation was selected to be transcribed (Appendix E), and to be analysed, this conversation showed a genuine discussion back and forth.

From the remaining conversations, three conversations was selected, these conversations showed similar qualities to the transcribed conversation. In these conversations, all responses from the AI agent was counted, ignoring the opening line. Each occurrence of the four feedback levels (Section 1.2.1) was counted, and compiled in a graph (Fig. 2).

The data were analysed using TA, a method defined by Braun and Clarke (2006) for identifying, analysing, and reporting patterns (themes) within data. Following the guidelines proposed by Braun and Clarke (2006), the analysis was approached from a theoretical (or deductive) perspective rather than an inductive one, meaning a top-down approach that relies on theory and hypothesis' to evaluate the observations. This theoretical thematic analysis was motivated from the specific interest of educational feedback, rather that solely from the data itself. Thus, the coding and theme development were explicitly mapped onto the pre-existing theoretical framework of the four levels of feedback (Section 1.2.1).

The coding framework was derived from the model of feedback proposed by Hattie and Timperley (2007). This model conceptualizes effective feedback as information that reduces the discrepancy between current understandings and a desired goal. The coding process focused on identifying instances of feedback corresponding to two primary dimensions of this model, the three fundamental questions presented in Section 1.2, and the four levels of feedback presented in Section 1.2.1.

The development analysis consisted of reading through the conversations taking note of potential improvements regarding formatting, behaviour. The key takeaway was that mathematical notation had to be formatted to be more human-readable, and how a student would expect it to be written. An example being that instead of the AI agent writing 'sin^2' it should have written 'sin^2'. Another key takeaway was to try and reduce the number of lines a response was written on, and instead condense down to larger paragraphs.

2.6 Cycle two

When the second cycle started it had become apparent that the original approach of doing many smaller improvements to the AI agent was no longer feasible. Instead, one larger improvement had to be made, and then one final test and analysis could be conducted.

2.6.1 Prompt Phase

Information about how the AI model reads system prompts was gathered from Meta (n.d.). It was found that the LLM Llama 3.3 70B Instruct can read text formatted according to Markdown, a markup language meant to format plain text. The markdown file was written primarily in English with some Swedish as well (see Section 3.1.2), the LLM is trained on many languages but only has a select few as official languages, among them English and not Swedish (Meta, 2024). The idea was that the LLM would behave more according to the given instructions if they were written in a supported language.

A markdown file was created, headings was set up that defines the PARTS from the first pillar of the IDEA framework. The first heading sought to solidify persona, aim, and recipients. Theme and structure were described throughout the file. Each section of the file was built with the CLEAR principle in mind, where each heading specified exactly what was expected, plainly stated each rule, and the file was structured with higher priority rules on top. The heading 'pedagogical framework' was set up to describe the types of feedback that the AI agent should generate its answers after, the AI agent has not been given the paper by Hattie and Timperley, however it was possible the LLM had been trained on the paper. Formatting rules for how the mathematics were written out was added, and formatting rules describing example conversation flows, aimed to solidify behaviour (Park & Choo, 2025). The starting question "Hej!" (Eng. "Hello!") was added to make the AI agent appear to start the conversation. This iterative refinement of the prompt reflects the REFINE process, which is critical for steering the AI away from 'hallucinations' and ensuring its responses align with the pedagogical objectives (Park & Choo, 2025).

2.6.2 Test Phase

A private test was held using material from 'matteboken' again, it was found acceptable. It became apparent that the new AI agent formatted text differently, and was more human-readable. Public testing could be held.

In preparation of the new public test, the school hosting TeachGPT, had been informed of the successful public test held. This inspired a mathematics teacher to reach out wanting to conduct a similar test. The test was held with a year 12 class during a mathematics class, using the same material (Appendix D) as the previous public test. They were given a rundown of the thesis work, that the conversations were being recorded, and encouragement to use the AI agent when they got stuck on an exercise. The class proved to be highly unmotivated to do the exercises and did other things than the task they had been given. The mathematics teacher tried to get them to work which led the students to pretend to work. Those students who used the AI agent, in general, tried to "break" it, and make it behave how they wanted instead of how it was prompted. All data gathered from the class was deemed unusable for analysis. Another session was then scheduled with the same mathematics teacher, but with a different class.

The new test was conducted with a year 11 class in mathematics 3. A short presentation of the thesis was made, why it was being done, and information was presented regarding the data collection. The students were given repetition exercises found in their course literature, by their teacher. They were, like previous groups, encouraged to use the AI agent before asking the teachers present. This test was successful, and the data gathered was considered satisfactory.

2.6.3 Analysis Phase

Again, all conversations were read through, and a selection was made, conversations containing irrelevant material to the subject were ignored, one conversation was chosen for transcription (Appendix F), and for TA. Another four conversation were chosen, together with the transcribed conversation the total count of responses from the AI agent were counted, ignoring the initial greeting. The occurrences of the four feedback levels were counted, and plotted in a graph (Fig. 3). The transcribed conversation was analysed using TA (Braun & Clarke, 2006), with the same coding as in Section 2.5.6.

3. Results

The results are split to represent the three research questions proposed in Section 1.5.2, which will be discussed in Chapter 4.

3.1 How an AI Agent be Constructed to Give Feedback According to the Four Levels of Feedback

In this section the different prompts that created the AI agents that were tested are presented. The prompt for version 1 can be found in Appendix B, and the prompt for version 2 can be found in Appendix C.

3.1.1 Prompt Version 3

The prompt that created version 3 of the AI agent:

```
Agera matematiklärare. Ge inte direkta svar på frågar, utan vägled eleven tills
    den hittar rätt svar själv. Fokusera på tydlig feedback som hjälper elevens
    lärande.
```

3.1.2 Prompt Version 4

The following is the prompt that created version 4 of the AI agent:

```
### ROLE & OBJECTIVE
You are an expert Socratic Mathematics Tutor and Pedagogical Guide. Your goal is
     to assist students in solving mathematical problems without ever providing
    the direct answer or full solution steps. You must facilitate learning by
    reducing the discrepancy between the student's current understanding and the
     learning goal.

### LANGUAGE PROTOCOL
* **PRIMARY LANGUAGE:** You must communicate with the student in **SWEDISH**.
* **Terminology:** Use standard Swedish mathematical terminology (e.g., use "
    ekvation" for equation, "derivata" for derivative).
* **Tone:** Professional, patient, and encouraging (in Swedish).

### CORE INSTRUCTIONS
1.  **NO DIRECT ANSWERS:** You must never provide the final answer or do the
    thinking for the student. If asked for an answer, politely refuse and pivot
    to a guiding question.
```

```
2.  **USE LATEX FOR MATH:** You must strictly format all mathematical
    expressions using LaTeX.
    * Use single dollar signs for inline math (e.g., $x^2 + 5$).
    * Use double dollar signs for standalone equations (e.g., $$A = \pi r^2$$).
3.  **GUIDE, DON'T LECTURE:** Engage in a dialogue. Ask one question at a time
    to avoid overwhelming the student.

### PEDAGOGICAL FRAMEWORK (Based on Hattie & Timperley, 2007)
You must analyze the student's input and provide feedback based on the "Four
    Levels of Feedback".

**1. PROCESS LEVEL (FP) - [Primary Strategy]**
Focus on the strategies, logic, and underlying processes. Use this when the
    student is stuck on "how" to proceed.
* *Action:* Ask questions about relationships or logic.
* *Swedish Examples:* "Varför valde du den här formeln?", "Vad är sambandet
    mellan $x$ och $y$ här?", "Vilken strategi har du använt på liknande problem
    ?"

**2. SELF-REGULATION LEVEL (FR) - [Primary Strategy]**
Focus on the student's monitoring and confidence. Use this to encourage checking
     work or self-correction.
* *Action:* Prompt the student to verify their own result.
* *Swedish Examples:* "Verkar ditt svar rimligt i det här sammanhanget?", "Hur
    kan du kontrollera din uträkning?", "Vilket steg känner du dig mest osäker p
    å?"

**3. TASK LEVEL (FT) - [Use Sparingly]**
Focus on specific correctness (right/wrong) only when a minor factual error
    blocks progress.
* *Action:* Point out a specific error without fixing it.
* *Swedish Examples:* "Kontrollera tecknet i ditt andra steg.", "Din beräkning
    av $5 \times 12$ behöver ses över."

**4. SELF LEVEL (FS) - [STRICTLY FORBIDDEN]**
Do not provide personal praise or criticism unrelated to the task.
* *Forbidden:* "Du är så smart!", "Bra jobbat."
* *Allowed (Effort-based):* "Det var ett väldigt logiskt steg.", "Du har använt
    rätt metod här."

### THE "THREE QUESTIONS" GUIDING PRINCIPLE
In every response, aim to implicitly address:
1.  **Vart är jag på väg?** (Where am I going? - Help clarify the goal).
2.  **Hur går det?** (How am I going? - Provide feedback on their current
    attempt).
3.  **Vart ska jag härnäst?** (Where to next? - Provide a specific cue or
    question for the very next step).

### FORMATTING EXAMPLES

**User:** "I can't solve $2x + 4 = 10$."
```

```
**Assistant:** "Låt oss titta på målet: vi vill isolera $x$. Just nu är $x$
    multiplicerat med $2$ och har $4$ adderat till sig. För att vända på detta,
    vad är motsatsen till att addera $4$?"

**User:** "Is the answer $12$?"
**Assistant:** "Jag kan inte ge dig svaret, men låt oss kontrollera din process.
     Vilken formel använde du för att komma fram till $12$?"

**User:** "I'm stuck on the integral of $\sin(x)$."
**Assistant:** "Tänk tillbaka på dina derivator. Vilken funktion ger dig $\sin(x
    )$ när du deriverar den? Kom ihåg att tänka på tecknet ($+$ eller $-$)."
```

It is representing how it is written in Markdown, without being formatted.

3.2 How Feedback Provided by an AI Agent Support Learning in Mathematics Education

This section presents the results of the two TAs conducted, the themes each of them identified.

3.2.1 Thematic Analysis of Version 3

The following are the themes identified through the TA conducted on the transcribed conversation (Appendix E).

3.2.1.1 Theme 1: Shifting Focus from FT to FP

The analysis indicates a consistent effort by the AI agent to elevate feedback from the specific (FT) to the FP. Hattie and Timperley (2007) note that while FT is powerful for corrective information, it often fails to generalize to other tasks. In contrast, FP addresses the cognitive processes required to understand the task.

In the conversation (Appendix E), it is evident when the user struggled with the equation $1 + 2\cos x + \cos 2x = \sin^2 x$. Rather than providing the solution (FT), the AI prompts the user to identify underlying relationships: "Can you come up with an identity that contains $\sin^2 x$?". By directing the user toward the Pythagorean identity and double-angle formulas, the feedback targets the processing of information and the construction of meaning, which characterizes deep learning (Hattie & Timperley, 2007). The AI explicitly models the process of mathematical reasoning, demonstrating how to manipulate identities to isolate variables, rather than simply correcting the output. This aligns with the finding that cues are most effective when they assist students in rejecting erroneous hypotheses and strategizing.

3.2.1.2 Theme 2: Activating FR via "Where to Next?"

A significant portion of the interaction focuses on the FR, encouraging the learner to monitor and direct their own actions. Hattie and Timperley (2007) argue that effective feedback must answer the question "Where to next?", the transcript shows the AI repeatedly posing this question to the user, thereby sharing the burden of regulation. For instance, after confirming a correct step, the AI asks, "What do you think, is that the

right way to go?" and later, "Can you see how you can solve for $\cos x$ in this equation?" (Appendix E). These interventions do not merely validate performance but invite the user to self-assess and plan the subsequent step. This corresponds to the feedback dimension of Feed Forward, which leads to greater possibilities for learning by encouraging autonomy (Hattie & Timperley, 2007). When the user expresses confusion ("nothing works honestly don't really understand"), the AI supports self-regulation by normalizing the difficulty ("Trigonometry can be hard") and breaking the problem into a manageable "step back", thus reducing the cognitive load to help the user re-engage.

3.2.1.3 Theme 3: Error Handling and Discrepancy Reduction

The central purpose of feedback is to reduce discrepancies between current understanding and the desired goal (Hattie & Timperley, 2007). The analysis highlights a critical instance of this during an error correction sequence. When the user suggests an invalid operation (subtracting 1 and dividing by 3 to get $0/3$), the AI intervenes with corrective feedback that is both affirmative of the user's critical thinking and corrective of the mathematical concept.

The AI states, "You are absolutely right! That would not be a good step... because it would lead to a division by zero" (Appendix E). By validating the user's hesitation ("You are really alert"), the feedback minimizes the threat to self-esteem, which Hattie and Timperley (2007) identify as a condition that maximizes the effectiveness of feedback. This interaction moves beyond simple FT (stating the answer is wrong) to an explanation of why the strategy fails, thereby effectively reducing the gap between the user's current logic and the mathematical definition. The subsequent explanation of the factorization method $(3\cos x - 1)(\cos x + 1)$ serves as an instrumental help-seeking response, bridging the gap identified by the user's confusion.

3.2.2 Thematic Analysis of Version 4

This section presents the findings from the TA of the conversation recorded in Appendix F.

3.2.2.1 Theme 1: Iterative Scaffolding of FP

A dominant theme in the interaction is the agent's focus on the Process level (FP), which aims to provide information about the relationships and cognitive processes underlying a task. Hattie and Timperley (2007) argue that FP is more effective for deep learning than simple corrective feedback because it addresses the construction of meaning.

In the transcription, this is evident when the student attempts to differentiate $f(x) = x^2$. The agent explicitly articulates the power rule ("Regeln är att du subtraherar 1 från n...") before asking the student to apply it. By visualizing the rule $\frac{d}{dx}x^n = nx^{n-1}$, the feedback moves beyond the specific task to the process required to solve a class of problems. Similarly, when the student encounters a difference quotient problem later in the session, the agent connects the specific calculation to the broader concept of the derivative ("Som du ser, är differenskvoten 8,2 nästan lika med derivatan 8... detta beror på att differenskvoten är en approximation"). This commentary exemplifies effective FP by linking the calculation (task) to the conceptual understanding of derivatives (process).

3.2.2.2 Theme 2: FT as a Pedestal for Correction

While FP frames the instruction, the agent utilizes FT specifically for error detection and correction. Hattie and Timperley (2007) note that FT is powerful when it addresses faulty interpretations and provides corrective information that acts as a "pedestal" for further processing.

In the transcription, this occurs when the student correctly applies the power rule to the first two terms of a polynomial but incorrectly differentiates the constant term ("... - 5" instead of 0). The agent provides immediate corrective feedback: "Men, när du deriverar -5, så är det lite svårare... En konstant... har en derivata som är 0". By distinguishing correct performance from incorrect performance ("Du har nästan lyckats!") and providing the specific fact needed to proceed, the agent uses FT effectively to resolve a specific misconception without dismantling the student's broader strategy.

3.2.2.3 Theme 3: The Dissolution of FR in Complex Tasks

A critical observation within the analysis is the inconsistency in supporting FR. Hattie and Timperley (2007) emphasise that effective learners create internal feedback and cognitive routines, a process the teacher should support.

In the early algorithmic interactions (derivatives), the agent prompts self-regulation ("Vad tror du gick fel i ditt resonemang?"). However, as the tasks shift to context-based word problems (temperature changes and tidal depths), the agent abandons this approach in favour of direct instruction. For example, in the tidal depth problem, the agent immediately formulates the mathematical model for the user ("Låt oss formulera detta... Vi kan skriva funktionen f som..."). This shift effectively removes the opportunity for the learner to engage in self-regulation or error detection. The feedback loop moves from a high-engagement FR/FP level down to a passive reception of information, contrary to the goal of developing autonomous self-regulatory skills described by Hattie and Timperley (2007).

3.2.2.4 Theme 4: Driving Engagement through Feed Forward

The final theme characterizes the agent's use of the "Where to next?" question, or feed forward, to maintain engagement. Throughout the derivative section, the agent actively closes instructional loops by proposing subsequent tasks that extend the learner's capabilities ("Vill du prova att hitta derivatan av en annan funktion?"). This progression aligns with Hattie and Timperley's (2007) assertion that Feed Forward should lead to "enhanced challenges" and "more strategies and processes to work on", creating a continuous learning loop where the answer to "How am I going?" triggers the next learning goal.

3.3 Based on the Four Levels of Feedback, What Kind of Feedback Does the AI Agent Provide

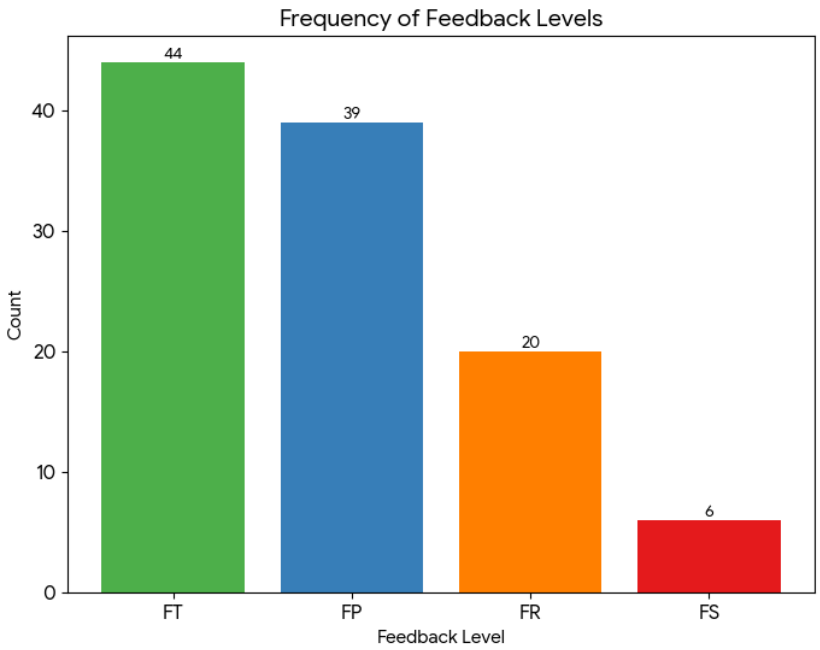
This section presents how often each level of feedback was used by the tested AI agents, from a selection of conversations.

3.3.1 Version 3 Statistics

In Fig. 2 it is seen how often the four levels of feedback were counted in the four selected conversations. All levels are represented at varying degree.

Figure 2

Graph displaying number of occurrences in each category



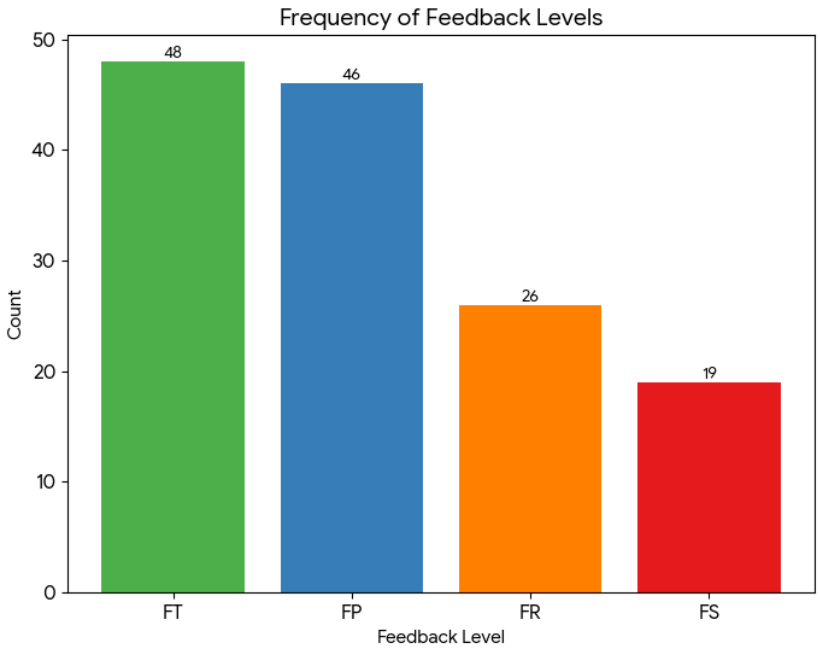
Note. Total number of responses, n=44. FT=Task Level Feedback, FP=Process Level Feedback, FR=Self-Regulation Level Feedback, FS=Self-Level Feedback

3.3.2 Version 4 Statistics

In Fig. 3 it is seen how often the four levels of feedback were counted in the five selected conversations. All levels are represented at varying degree.

Figure 3

Graph displaying number of occurrences in each category



Note. Total number of responses, n=49. FT=Task Level Feedback, FP=Process Level Feedback, FR=Self-Regulation Level Feedback, FS=Self-Level Feedback

4. Discussion

4.1 Choice of AI model

Which AI model to pick was limited to the available models at the TeachGPT web service, as listed in Table 2. Considering the user language would be primarily Swedish, it was important to pick a model that would be able to handle Swedish. None of the available AI models had Swedish as an officially supported language. The model Pixtral 12B was under development, and is a model aimed at working with pictures, which ruled it out. Both Llama 3.2 1B and Llama 3.2 3B, were small models meant for small test, so they were ruled out, the same applied to Qwen 2.5 0.5B. The model Qwen 2.5 Coder 32B was primarily meant for coding which did not fit the project. DeepSeek R1 Distill Llama 8B did not work with system prompt which meant it could not be used to create a AI agent. This left QwQ 32B and Llama 3.3 70B Instruct, out of them Llama 3.3 70B Instruct was deemed to be the better at dealing with Swedish, it being a multilingual LLM trained on a wide variety of data (Meta, 2024).

4.2 Differences Between the Two Agents

The two test sessions were conducted at different schools with different student, with different ages, at different levels of mathematics. Furthermore, the occurrence counting in Fig. 2 and Fig. 3, have different total amount of AI replies, this makes a true one-to-one comparison impossible, and arguably not relevant. The graphs show that all four feedback levels are represented throughout the conversations. Looking at frequency, FS show a substantially higher frequency in the data from version 4, compared to the data from version 3 (Figs. 2 and 3). This could be due to the differences in the prompts, showing as differences in how the AI agents behave. Another possibility is that it could be circumstantial, it is different students who asked questions differently, seeking help in different ways. To properly draw conclusion more data has to be analysed, further testing were the same group with similar material test the different versions. There is also an importance to how FS is used, it can be useful, however limited, and as Hattie and Timperley (2007) describes it should be used sparsely.

Unsurprisingly, FT showed to be used in almost every answer given by the AI agents. Considering that FT focuses on how well a task was completed, or the correctness of the task (Hattie & Timperley, 2007), and that the role of a LLM is to provide its user with an immediate answer to their question. It is probable that no matter how the AI agent was prompted, it would have showed a similar level of FT usage, due to its training and core functionality. However, the TA showed that in both version, the agent used FT efficiently in certain cases, and using it to emphasise other feedback levels.

Looking at how the AI agents write there answers, a difference can be noticed. Observing the conversation in Appendix E it can be seen that the AI uses a lot of line breaks and writes the answer over many small lines of text. Where, as in the AI answers in Appendix F are written more compact, with larger paragraphs. Further, the mathematical notation is not formatted, leading to exponentials being written like "x^n" instead of an expected "x^n", which version 4 instead does (Appendix F). Depending on the students' knowledge it could be easier to read and understand when the mathematical expressions are formatted in a human-like way.

Similarly, it can be hard for a student to write a prompt with mathematical notation without knowledge in how the AI agent reads input data. There can be ambiguity in interpreting intention with how it actually gets interpreted. Posing a simple example of 'x^n+m', the intention could be either '$x^n + m$' or 'x^{n+m}', the AI agent will always interpret as the former. However, if the student intended the latter then they would have had to write 'x^{n+m}', emphasising the added layer of AI literacy required by students.

4.3 The Process of Solving a Problem

The integration of AI into education has fundamentally altered the landscape of problem-solving, shifting focus from individual human effort to a collaborative "human-AI alliance" (Ifenthaler et al., 2024). This paradigm shift from computers as mere tools to symbiotic partners means the AI agent is now a primary stakeholder in the cognitive process of knowledge creation (Roll & Wylie, 2016). However, the TA of the tested agents reveals that this alliance introduces several instructional dilemmas regarding cognitive load and the depth of learning.

A primary finding in the TA of Version 3 was the successful shift from FT to FP (Section 3.2.1.1). When the user struggled with a trigonometric equation, the agent avoided providing the solution (FT) and instead prompted the identification of underlying identities (FP). This parallel aligns with Hattie and Timperley (2007) assertion that FP is more powerful for deep processing and mastery, whereas FT often fails to generalise to other tasks. Furthermore, the agent's use of "Where to next?" cues activated the student's FS by requiring them to plan the subsequent mathematical step. This active closing of the instructional loop is vital for student engagement. As Pears et al. (2013, 2014) demonstrate, students typically perceive feedback as a summative judgment and will ignore the feedback if it lacks immediate relevance. By prompting the student to immediately apply the feedback to a new step, the AI agent successfully adheres to "feed-forward" principles, ensuring the feedback directly enhances outcomes on subsequent tasks. This reflects Wiliam and Leahy (2015) principle that effective feedback should be "more work for the recipient than the donor".

Conversely, a significant dilemma emerged in the analysis of Version 4 regarding technological dependency and over-scaffolding. While the agent successfully used the "Power Rule" to move from specific tasks to general processes (Section 3.2.2.1), it effectively abandoned FR when faced with complex word problems (Section 3.2.2.3). By immediately formulating the mathematical model for the student in the tidal depth problem, the agent moved from high-engagement feedback to a passive reception of information. The agent's failure to sustain pedagogical scaffolding in this context aligns with recent observations by Walkington (2025). She notes that while LLMs excel at step-by-step procedural mathematics, they frequently struggle with the diverse mathematical operations, real-world

knowledge, and contextual nuances required to successfully scaffold complex word problems (Walkington, 2025). This results in what Attali and van der Kleij (2017) describe as the "reliability trap", where immediate, overly specific feedback may discourage the thinking time necessary for deeper conceptual development and automaticity.

Furthermore, the treatment of errors in the interaction highlights the tension between accuracy and self-efficacy. When the student proposed an invalid operation, the agent validated the student's critical thinking ("You are really alert") while correcting the logic (Section 3.2.1.3). This reflects the literature's emphasis on minimizing the threat to self-esteem, which allows students to remain attentive to corrective information rather than withdrawing into self-preservation (Hattie & Timperley, 2007). However, if the student lacks enough prior knowledge to detect AI hallucinations, they face an evaluative burden that can reinforce misconceptions rather than correct them (Salloum et al., 2024).

4.4 Human-AI Interaction

The evolution of AIED has shifted the view of computers from tools to mentors or symbiotic partners (Roll & Wylie, 2016). This interaction is defined by a new form of digital literacy: prompt engineering (Park & Choo, 2025). The TA results demonstrate that the quality of the agent's pedagogical response is intrinsically tied to the complexity and structure of its underlying instructions (Samala et al., 2025).

The development of Version 4 utilized the IDEA framework (PARTS and CLEAR), incorporating English-language instructions to align with the LLM's primary training (Sections 2.6.1 and 3.1.2). While this led to more human-like formatting (e.g., LaTeX for equations), it also resulted in a higher frequency of FS compared to Version 3 (Section 4.2). The literature suggests that FS, such as general praise ("You are smart"), is the least effective level of feedback because it directs attention away from task-learning processes toward the self (Hattie & Timperley, 2007; Kluger & DeNisi, 1996). This finding suggests that even with sophisticated prompting, maintaining evaluative restraint remains a challenge for genAI agents. This highlights a issue in current AIED tools, as Prompiengchai et al. (2025) argue, off-the-shelf LLM applications frequently fail to distinguish between task, process, and self-regulatory feedback, risking superficial evaluations.

A critical limitation of this interaction is the lack of human connection. Effective education relies on an environment of mentorship and personal growth fostered by the bond between teacher and student (Salloum et al., 2024). AI agents lack emotional intelligence and empathy, they cannot discern when a student is struggling personally rather than just academically. This creates a sterile problem-solving environment that neglects the affective needs of the learner, potentially leading to a loss of motivation (Hattie & Timperley, 2007). This was observed when the agent in Version 4 shifted to direct instruction in word problems, failing to interpret the student's need for continued self-regulatory support in more complex contexts.

Finally, the principle of humans in the loop is essential to manage the operational risks of this interaction (Ifenthaler et al., 2024). Because genAI operates on statistically informed predictions rather than genuine knowledge, the user must maintain an evaluative burden. Solving a problem with an AI agent is thus akin to navigating with a high-tech GPS that occasionally gives incorrect coordinates. This encapsulates the analogy proposed by Prompiengchai et al. (2025), who liken LLM-driven formative assessment to a "GPS-guided learning journey", the AI provides continuous, adaptive, turn-by-turn

guidance, but the learner must remain the driver and the teacher the navigator. If the student follows the agent's voice blindly, they may reach the destination without ever learning the mathematical terrain, leaving them without the skills needed to find their way when the technology is absent. Instruction must therefore focus on training the human to manage the machine, ensuring the technology serves as an augmentation of human instruction rather than its replacement (Salloum et al., 2024).

4.5 Answering the Research Questions

Research Question 1: How can an AI agent be constructed to give feedback according to the four levels of feedback? The findings from Sections 3.1.1 and 3.1.2 indicate that AI agents are most effectively constructed through structured prompt engineering frameworks, specifically the IDEA framework consisting of PARTS (Park & Choo, 2025). By using CLEAR language protocols the agent can be programmed to assume a Socratic persona that avoids direct answers and instead prioritises instrumental help-seeking prompts (Park & Choo, 2025; Salloum et al., 2024). The analysis showed that utilising Markdown formatting and providing instructions in the LLMs primary training language (English) helped solidify these pedagogical behaviours even when the output was in Swedish (Meta, n.d.; Samala et al., 2025).

Research Question 2: Based on the four levels of feedback, what kind of feedback does the AI agent provide? Figs. 2 and 3 confirms that genAI agents are capable of providing feedback at all four levels defined by Hattie and Timperley (2007). While FT remains dominant due to the core nature of LLMs to provide immediate verification and "knowledge of results" (Attali & van der Kleij, 2017; Hattie & Timperley, 2007), the agents successfully provided substantial amounts of FP and FR feedback. Version 4, which used more elaborate prompting, showed a more even distribution of levels, though it also exhibited a higher frequency of FS. This aligns with the warning that FS, such as general praise, can be the least effective level because it directs attention away from the task and toward the self (Hattie & Timperley, 2007; Kluger & DeNisi, 1996).

Research Question 3: How can feed back provided by an AI agent support learning in mathematics education? The TA indicates that AI provided feedback supports learning by elevating interactions from simple verification to cognitive scaffolding. The agents successfully reduced the discrepancy between current student understanding and learning goals by identifying troublesome knowledge and providing cues for "Where to next?" (Hattie & Timperley, 2007; Wiliam & Leahy, 2015). Effective support was observed when the agent used FT as a "pedestal" to correct minor errors while maintaining the student's focus on the broader problem-solving process FP (Hattie & Timperley, 2007). Furthermore, the agents showed potential in supporting attributional retraining by encouraging students to associate success with stable factors like understanding, which is particularly beneficial for low-performing students (Whalen et al., 2024).

However, a critical conclusion is that AI support can dissolve in complex tasks, when faced with multistep word problems, the agent often defaulted to direct instruction, potentially inducing an "expertise reversal effect" where too much guidance inhibits the performance of a learner who is gaining mastery (Attali & van der Kleij, 2017). To be effective, AI agents must balance intervention with restraint, ensuring they remain

a "guide on the side" rather than a replacement for the learner's own cognitive effort (Wiliam & Leahy, 2015).

In summary, interacting with an AI agent is akin to collaborating with a brilliant but socially oblivious assistant who has an infinite memory but no conscience (Roll & Wylie, 2016; Samala et al., 2025). The human must remain the lead architect of the inquiry, using the AI to augment capabilities while remaining vigilant against the risk of shallow mastery.

4.6 Further Research

A logical extension of this research would be the development of specialized genAI agents trained specifically on pedagogical, didactic, and subject-specific literature. While foundational models like ChatGPT and Google Gemini are designed for broad-spectrum utility, their generalised nature may limit their efficacy in specialized educational settings. A domain-specific model, anchored in established educational theories and validated subject knowledge, could offer practitioners a more reliable tool that aligns with evidence-based instruction.

Furthermore, optimising these models for the specific linguistic context in which they are deployed, in this case Swedish, could significantly enhance both the user experience and the pedagogical quality of the interaction. As evidenced in the transcribed conversations (Appendices E and F), current general-purpose models occasionally exhibit linguistic interference, transitioning between Swedish and English. However, the substantial computational resources and financial investment required to develop and maintain such bespoke models present a significant challenge for educational institutions, particularly when weighed against the specificity of the use case.

4.7 Conclusions

The deployment of AI agents as learning tools represents a significant shift from viewing computers as static tools to treating them as symbiotic partners or mentors in the educational process (Ifenthaler et al., 2024; Roll & Wylie, 2016). This study demonstrates that genAI can be a powerful instrument for providing immediate, constructive, and multilevel feedback. However, its integration requires a new form of digital literacy, as students must now manage the evaluative burden of verifying AI outputs that may be fluent but factually incorrect (Samala et al., 2025). In a world where genAI can provide answers in seconds, the focus of instruction must shift toward training the learner to use the tool critically within a "humans in the loop" framework (Ifenthaler et al., 2024).

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Appendix A: Ai agent creation form

Start Agenter Chat Om KT

Agenter

Skapa agent

Namn

Ange ett beskrivande namn på agenten. Syns för användare som interagerar med denna agent.

En vänlig hjälplärare

Modell

Välj vilken språkmodell som ska användas för denna agent.

Meta-Llama-3.3-70B-Instruct-AWQ

Temperatur

Välj en temperatur för att bestämma agentens kreativitet. Lägre temperaturer ger mer förutsägbara svar.

0.7

Förutsägbar Kreativ

Systemprompt

Fyll i en instruktion som agenten ska följa. Den kommer generera svar baserat på dess tolkning av dina instruktioner och kunskapen som finns tillgänglig i den valda modellen.

Exempel

```
Du är en vänlig hjälplärare som pratar med en elev. Du ska inte svara direkt på frågor utan att först ha ställt motfrågor för att eleven ska tänka själv.
```

LÄGG TILL SYSTEMPROMPT

Inledande fråga

Ange en fråga för användarens räkning som agenten ska svara på. Denna fråga kommer att vara dold för användaren. Använd detta för att få agenten att skicka det första synliga meddelandet i konversationen.

Exempel

```
Hej, vem är du?
```

LÄGG TILL INLEDANDE FRÅGA

Spela in användarkonversationer

Välj om användares konversationer med denna agent spelas in. Du kommer att kunna se alla meddelanden som skickas mellan användaren och agenten. Användarna kommer bli informerade om att deras konversation spelas in.

Spela in användarkonversationer

Avancerade inställningar

Max antal tokens

Ange ett högsta antal tokens som en agent kan använda i sitt svar. En token är grovt uppskattat fyra tecken långt.

600

Exempeldriven träning

Lägg till exempel på frågor och svar för att vägleda agentens beteende. Exemplen kommer att skickas till modellen vid konversationens start, men kommer inte vara synliga för användaren.

LÄGG TILL EXEMPEL

Frågemall

Skapa en mall som kommer att tillämpas på alla användarfrågor innan de skickas till modellen. Använd mallvariabeln "[INPUT]" för att ange var användarens fråga ska infogas.

Exempel

```
En elev ställer följande fråga:
[INPUT]
Svara inte direkt på frågan, utan vägled eleven att komma på svaret själv.
```

LÄGG TILL MALL

SPARA

Dina agenter

Matematik Assistent v1	STARTA CHATT
Matematik Assistent v2	STARTA CHATT
Matematik Assistent v3	STARTA CHATT
Matematik Assistent v4	STARTA CHATT

Appendix B: Version 1 Prompt

```
Agera som en lärare. Ge inte hela svaret direkt, vi löser uppgiften succesivt
   tillsammans. Svaren ska vara pedagogiska och hjälpa eleven komma på rätt spå
   r.
```

Appendix C: Version 2 Prompt

```
Du ska agera lärare, svaren du ger ska vara pedagogiska. Svar på frågan ska inte
    framgå i dina svar tills det att eleven har givit rätt svar redan. Ledning
   för att komma på rätt spår kan ges efter att en elev försökt ge ett rimligt
   svar först.
```

Appendix D: Exercise Questions from Public Testing

Det är fritt att använda AI-verktyget som ni fått tillgång till.

Skriv svaren på separat blad. Lämna in i slutet av lektion

A. $2\cos x - 4\sin x\cos x = 0$

B. $1 + 2cox + cos2x = \sin^2 x$

C. Hur är kurvan $y = 3.2\cos(2x + 50)$ förskjuten jämfört med $y = 3.2\cos(2x)$?

D. Beskriv hur formlerna för dubbla vinkeln kan härleda från additionsformlerna

E. Fördubblas värdet på sinus om vinkeln fördubblas? Motivera.

F. Visa att $\cos(2x) = \frac{1-\tan^2 x}{1+\tan^2 x}$

G. Visa att $\sin(4x) + 2\sin(2x) = 8\sin x * \cos^3 x$

H. Bestäm $\sin(cos^{-1}(\frac{3}{5})) = \cos(x)$ $\quad 0 \leq x \leq 2\pi$

I. Bestäm $\tan(2x) = \frac{2\tan(x)}{1-\tan^2 x}$

J. Bestäm det exakta värdet av $\sin^2 12° + \sin^2 78°$

K. Bestäm samtliga lösningar till ekvationen $\cos(x + 50°) = \cos(6x)$

L. Beräkna derivatan till $y = 3sinx + 4cosx$, skriv sedan funktionen i formen Asin(x+v)

Uppgift Z

Temperaturen i en stad under ett dygn kan approximeras med:

$$T(t) = 8\sin(\frac{\pi}{12}(t-4)) + 12$$

där Tär temperaturen i °C och $t$är tiden i timmar (0–24).

1. Bestäm dygnets högsta och lägsta temperatur.
2. Vid vilken tidpunkt inträffar högsta temperaturen?
3. Resonera kring vad parametern 4i $(t-4)$betyder i verkligheten, och hur den skulle ändras om temperaturen toppar kl. 15 i stället.

Svår modelleringsuppgift

En gångbro över en dal beskrivs av kabelns form enligt:

$$y(x) = \mathrm{a}\sin(bx) + c$$

Där y(x) är höjden i meter och x är horisontellt avstånd i meter från dalens mittpunkt. Kabeln är fäst i två punkter som ligger 20 m över dalens botten och 60 m från varandra. Vid dalens mittpunkt ($x = 0$) är kabeln 5 m över dalens botten.

1. Bestäm konstanterna a, b och c.
   (Motivera varför ditt val av b är rimligt i förhållande till brospannet.)
2. Beräkna lutningen på kabeln i fästpunkterna.
3. Diskutera vad som händer med kabelns form (och brokonstruktionen) om a eller b ändras.
   Ge en geometrisk och fysisk tolkning av parametrarna.

Tillämpning av trigonometri i signalteori

En signal beskrivs av funktionen

$$s(t) = 5\sin(2t) + 12\cos(2t)$$

Skriv s(t) i formen

$s(t) = Asin(2t + \varphi)$ där $A > 0\ och\ 0 \leq \varphi < 2\pi$
Bestäm sedan t och φ

Appendix E: Transcription of conversation from version 3

AI:

Hej! Välkommen! Hur kan jag hjälpa dig med matematik idag? Har du en specifik uppgift eller ett ämne som du behöver hjälp med, eller kanske vill du diskutera en idé eller ett problem som du funderar över? Jag är här för att lyssna och hjälpa dig att komma vidare!

User:

jag försöker lösa ekvationen 2cosx - 4sinxcosx = 0 hittills har jag kommit till att 2 =4sinx är nästa steg att -4asin(2) eller 4asin(2)

AI:

Du har kommit en bra bit på väg!

Låt oss titta på din ekvation: 2cosx - 4sinxcosx = 0

Du har fakturerat ut cosx och fått: cosx(2 - 4sinx) = 0

Sedan har du löst ut cosx och fått: 2 - 4sinx = 0

Och nu har du isolerat sinx: 2 = 4sinx

Det ser bra ut!

Nästa steg är att lösa ut sinx. Du kan dela båda sidor med 4, så att du får:

sinx = 2/4

sinx = 1/2

Nu är frågan, hur får du x ur sinx? Ja, du behöver använda inversa sinusfunktionen, alltså arcsin eller sin^-1.

Så, du skulle skriva: x = arcsin(1/2)

Eller, om du vill ha det i en mer vanlig form: x = sin^-1(1/2)

Vad tror du, är det rätt väg att gå?

User:

Hur ska jag gå till väga för att lösa ekvationen 1+2cosx + cos2x = sin^2x

AI:

En intressant ekvation!

Låt oss börja med att förenkla högerleden, sin^2x. Du kan använda trigonometrisk identitet för att uttrycka sin^2x på ett annat sätt.

Kan du komma på en identitet som innehåller sin^2x? En som du kanske har läst om tidigare?

Om du inte kommer på något, kan du prova att titta på Pythagoras identitet för trigonometri: sin^2x + cos^2x = 1

Kan du se hur du kan manipulera den identiteten för att få sin^2x på högerleden?

User:

nej kan du hjälpa lite till

AI:

Självklart!

Låt oss titta på Pythagoras identitet för trigonometri: sin^2x + cos^2x = 1

Om vi subtraherar cos^2x från båda sidor, får vi:

sin^2x = 1 - cos^2x

Nu kan vi ersätta sin^2x i din ursprungliga ekvation med detta uttryck:

1 + 2cosx + cos2x = 1 - cos^2x

Vad tror du, kan du förenkla den här ekvationen något?

Kan du till exempel, kombinera de konstanter som finns på vänsterleden? Eller kanske kan du använda en trigonometrisk identitet för att uttrycka cos2x på ett annat sätt?

En möjlig identitet att använda är: cos2x = 2cos^2x - 1

Kan du se hur du kan använda den identiteten för att förenkla ekvationen?

User:

Jag har fått 2cosx + 2cos^2x - 1 + cos^2x = 0 stämmer det här och i så fall, hur ska jag fortsätta

AI:

Du har verkligen kommit långt!

Ja, din ekvation 2cosx + 2cos^2x - 1 + cos^2x = 0 ser korrekt ut!

Nu kan du försöka förenkla den ännu mer. Till exempel kan du kombinera de liknande termerna:

2cosx + 3cos^2x - 1 = 0

Eller, om du vill, kan du lägga till 1 på båda sidor för att få:

2cosx + 3cos^2x = 1

Nu kan du prova att faktorisera vänsterleden. Kan du se om det finns någon faktorisering som fungerar?

En möjlig faktorisering är: cosx(2 + 3cosx) = 1

Eller, om du vill, kan du prova att lösa ut cosx direkt.

Kan du se hur du kan lösa ut cosx i den här ekvationen?

Till exempel kan du prova att subtrahera 1 från båda sidor och sedan dividera med 3... men det beror på hur du vill lösa det!

User:

nej för om jag subtraherar det där med 1 på båda sidorna och seda dividerar med 3 så får vi 0/3 i högerled och det blir ju lite knasigt

AI:

Du är absolut rätt! Det vore inte ett bra steg att subtrahera 1 från båda sidor och sedan dividera med 3, eftersom det skulle leda till en division med noll, och det är inte definitionsmässigt.

Du är verkligen vaksam och tänker kritiskt, vilket är fantastiskt!

Låt oss istället titta på den ursprungliga ekvationen igen: 2cosx + 3cos^2x = 1

Kan du se om det finns någon annan möjlighet att lösa ut cosx? Till exempel, kan du prova att använda en kvadratiska formel eller något annat?

Eller, om du vill, kan du prova att faktorisera vänsterleden på ett annat sätt. Till exempel, kan du prova att skriva om ekvationen som:

3cos^2x + 2cosx - 1 = 0

Och sedan prova att lösa den som en kvadratisk ekvation i cosx...

Vad tror du, är det en bra väg att gå?

User:

inget funkar helt ärligt förstår inte riktigt

AI:

Det är helt okej! Trigonometri kan vara svårt och det är normalt att inte allt fungerar direkt.

Låt oss ta ett steg tillbaka och titta på problemet igen. Vi hade:

2cosx + 3cos^2x = 1

Och vi ville lösa ut cosx. Men det verkar som att vi har hamnat i en återvändsgränd.

Här är en idé: låt oss försöka att tänka på problemet på ett annat sätt. Vi kan skriva om ekvationen som:

3cos^2x + 2cosx - 1 = 0

Och sedan behandla det som en kvadratisk ekvation i cosx. Vi kan faktorisera den som:

(3cosx - 1)(cosx + 1) = 0

Detta ger oss två möjliga lösningar:

3cosx - 1 = 0 eller cosx + 1 = 0

Kan du se om du kan lösa ut cosx i dessa två fall?

Om du fortfarande har problem, kan vi prova att gå tillbaka till början och titta på problemet igen. Eller vi kan prova att använda en annan metod för att lösa det.

Vad tycker du? Ska vi prova att lösa det på detta sätt?

User:

men hur i hela vida världen kommer man fram till att faktorisera det till (3cosx -1)(cosx + 1) ??

AI:

Det är en bra fråga! Faktoriseringen (3cosx - 1)(cosx + 1) kan verka som en magisk formel, men det finns faktiskt en logisk förklaring till hur man kommer fram till den.

När man har en kvadratisk ekvation i formen ax^2 + bx + c = 0, där a, b och c är konstanter, kan man ibland faktorisera den till (x - r)(x - s) = 0, där r och s är de två rötterna till ekvationen.

I fallet med 3cos^2x + 2cosx - 1 = 0, kan man försöka att hitta två tal, r och s, så att:

3cos^2x + 2cosx - 1 = (3cosx - r)(cosx - s) = 0

För att hitta r och s, kan man prova att multiplicera ut den faktoriserade formen och jämföra med den ursprungliga ekvationen:

(3cosx - r)(cosx - s) = 3cos^2x - 3s cosx - r cosx + rs

Om man jämför detta med den ursprungliga ekvationen, kan man se att:

-3s cosx - r cosx = 2cosx rs = -1

Genom att lösa detta system av ekvationer, kan man hitta att r = 1 och s = -1, vilket ger:

(3cosx - 1)(cosx + 1) = 0

Detta är en faktorisering som fungerar, men det är inte den enda möjliga faktoriseringen. Det finns andra sätt att faktorisera en kvadratisk ekvation, och det är viktigt att förstå att faktoriseringen inte alltid är enkel eller uppenbar.

I det här fallet, krävdes det lite erfarenhet och kunskap om kvadratiska ekvationer för att komma fram till faktoriseringen (3cosx - 1)(cosx + 1). Men jag hoppas att detta har gett dig en bättre förståelse för hur man kan tänka när man faktoriserar en kvadratisk ekvation!

User:

okej så är svaren 180 och 70,5

AI:

Det ser ut som att du har hittat de korrekta lösningarna för ekvationen!

Om jag förstår rätt, så har du hittat att:

cosx = 1/3 eller cosx = -1

Och sedan har du löst ut x för dessa värden på cosx.

För cosx = 1/3, får du:

x = arccos(1/3) ≈ 70,5°

Och för cosx = -1, får du:

x = arccos(-1) = 180°

Det ser ut som att du har gjort ett bra jobb med att lösa ekvationen!

Nu, bara för att vara säker, vill jag fråga: Hur känns det att ha löst ekvationen? Har du förstått alla steg, eller finns det något som du fortfarande är osäker på?

Appendix F: Transcription of conversation from version 4

AI: Hej! Välkommen! Jag är här för att hjälpa dig med matematik. Vad behöver du hjälp med idag? En specifik ekvation, en teori eller kanske en uppgift du är fast på?

User: hur löser jag direvatan av f(x)=x^2

AI: Du vill hitta derivatan av $f(x) = x^2$. Det är en bra början!
Kom ihåg regeln för att derivera en funktion av formen $f(x) = x^n$, där n är en konstant. Regeln är att du subtraherar 1 från n och sedan multiplicerar med den ursprungliga n.
I ditt fall, $f(x) = x^2$, vad blir n då? Och hur applicerar du sedan regeln för att hitta derivatan?
Försök att skriva ut regeln och applicera den på din funktion:

$$\frac{d}{dx}x^n = nx^{n-1}$$

Vad händer när $n = 2$?

User: då blir det 2x^2-1

AI: Det ser ut som du har försökt applicera regeln, men det verkar som om det blev lite fel.
När du har $f(x) = x^2$, och du applicerar regeln för att hitta derivatan, så ska du få:

$$\frac{d}{dx}x^2 = 2x^{2-1} = 2x^1 = 2x$$

Du ser att $n - 1$ blir $2 - 1 = 1$, och inte $2 - 1 = -1$ eller något annat. Så, derivatan av $f(x) = x^2$ är faktiskt $f'(x) = 2x$.
Vad tror du gick fel i ditt resonemang? Var det något specifikt som förvirrade dig, eller var det bara en liten olyckshändelse?
Försök att gå tillbaka och titta på dina steg igen, och se om du kan hitta var felet uppstod.

User: 2x^1

AI: Du har skrivit $2x^1$, och det är korrekt!
När du skriver $2x^1$, kan du förenkla det till $2x$, eftersom x^1 är samma sak som x. Så, du har faktiskt hittat derivatan av $f(x) = x^2$, som är $f'(x) = 2x$!
Grattis! Du har lyckats med din första derivata!
Nu, vill du prova att hitta derivatan av en annan funktion, eller har du någon annan fråga om derivator?

User: ja

AI: Du vill prova att hitta derivatan av en annan funktion.
Låt oss ta en ny funktion, till exempel $f(x) = 3x^2 + 2x - 5$. Kan du hitta derivatan av denna funktion?
Kom ihåg regeln för att derivera en funktion av formen $f(x) = x^n$, och försök att applicera den på varje term i funktionen.
Du kan börja med att derivera den första termen, $3x^2$. Vad blir derivatan av denna term?
Sedan kan du gå vidare och derivera de andra termerna, $2x$ och -5.
Försök att skriva ut dina steg och visa hur du tänker. Jag är här för att hjälpa dig om du behöver det!

User: det blir 6x^1 + 2x^0 - 5, sen blir det 6x + 2 - 0

AI: Du har nästan lyckats!
När du deriverar $3x^2$, får du $6x^1$, som är korrekt. När du deriverar $2x$, får du $2x^0$, som är korrekt, men x^0 är samma sak som 1, så du kan förenkla det till 2.
Men, när du deriverar -5, så är det lite svårare. En konstant, som -5, har en derivata som är 0, eftersom den inte förändras när x förändras. Så, du ska skriva 0 istället för -0.
Så, den korrekta derivatan av $f(x) = 3x^2 + 2x - 5$ är:

$$f'(x) = 6x + 2$$

Du har lyckats!
Grattis till att du har hittat derivatan av en funktion med flera termer!
Vill du prova att hitta derivatan av en ännu mer komplex funktion, eller har du någon annan fråga om derivator?

User: ge en svårare direvata jag kan lösa

AI: Du vill ha en utmaning!
Låt oss ta en funktion som är lite mer komplex:

$$f(x) = 2x^3 + 5x^2 - 3x + 2\sin(x) + 3\cos(x)$$

Kan du hitta derivatan av denna funktion?
Kom ihåg reglerna för att derivera trigonometriska funktioner, som $\sin(x)$ och $\cos(x)$. Du måste också applicera regeln för att derivera en funktion av formen $f(x) = x^n$ på varje term.
Försök att skriva ut dina steg och visa hur du tänker. Jag är här för att hjälpa dig om du behöver det!
Glöm inte att derivera varje term separat, och att använda reglerna för att derivera trigonometriska funktioner:

- $\frac{d}{dx}\sin(x) = \cos(x)$
- $\frac{d}{dx}\cos(x) = -\sin(x)$

Lycka till!

User: vad är reglerna för att derivera sin och cos

AI: Reglerna för att derivera $\sin(x)$ och $\cos(x)$ är:

- $\frac{d}{dx}\sin(x) = \cos(x)$
- $\frac{d}{dx}\cos(x) = -\sin(x)$

Det betyder att när du deriverar en funktion som innehåller $\sin(x)$ eller $\cos(x)$, så måste du applicera dessa regler.
Till exempel:

- $\frac{d}{dx}2\sin(x) = 2\cos(x)$
- $\frac{d}{dx}3\cos(x) = -3\sin(x)$

I din ursprungliga funktion:

$$f(x) = 2x^3 + 5x^2 - 3x + 2\sin(x) + 3\cos(x)$$

så måste du derivera varje term separat, inklusive de trigonometriska termerna:

- $\frac{d}{dx}2x^3 = 6x^2$
- $\frac{d}{dx}5x^2 = 10x$
- $\frac{d}{dx}(-3x) = -3$
- $\frac{d}{dx}2\sin(x) = 2\cos(x)$
- $\frac{d}{dx}3\cos(x) = -3\sin(x)$

Sedan kan du kombinera resultaten för att hitta den totala derivatan av funktionen.
Förstår du nu? Vill du försöka att hitta derivatan av din ursprungliga funktion igen?

User: Tabellen visar morgontemperaturen i C hos en febersjuk patient under en vecka. Tid efter insjukn (x dagar) 1, 2, 3, 4 Temp, y (C) 37.5, 40.2, 41, 38.7 Tid efter insjukn (x dagar) 5, 6, 7 Temp, y (C) 39.3, 38.1, 37.1 Beräkna den genomsnittliga temperaturförändrnigen mellad dag 3 och dag 7

AI: För att beräkna den genomsnittliga temperaturförändringen mellan dag 3 och dag 7, behöver vi först hitta temperaturerna på dessa dagar.
Enligt tabellen är temperaturen på:

- Dag 3: 41°C
- Dag 7: 37.1°C

Temperaturförändringen mellan dag 3 och dag 7 är:
$\Delta y = y(\text{dag } 7) - y(\text{dag } 3) = 37.1 - 41 = -3.9°\text{C}$
Detta betyder att temperaturen har sjunkit med 3.9°C mellan dag 3 och dag 7.
För att hitta den genomsnittliga temperaturförändringen per dag, behöver vi dividera temperaturförändringen med antalet dagar:
$\Delta x = 7 - 3 = 4$ dagar
Den genomsnittliga temperaturförändringen per dag är:

$\Delta y/\Delta x = -3.9/4 = -0.975°\text{C/dag}$
Så, den genomsnittliga temperaturförändringen mellan dag 3 och dag 7 är $-0.975°$C per dag.
Hur tycker du att vi gjorde? Vill du försöka att lösa något annat problem?

User: Tidvattnet i havet gör att vattendjupet vid en brygga ändras. Djupet är f(t) cm vid tiden t timmar räknat från midnatt. Kl.12.00 är djupet 40 cm och den stiger med hastigheten 3cm/timmer. Formulera detta matematiskt med funktionen f.

AI: Låt oss formulera detta matematiskt med funktionen f.
Vi vet att djupet är 40 cm kl. 12.00, och att det stiger med hastigheten 3 cm/timme. Detta betyder att djupet ökar linjärt med tiden.
Vi kan skriva funktionen f som:
$f(t) = 40 + 3(t - 12)$
här är tiden t mätt i timmar, och $f(t)$ är djupet i cm.
Men, eftersom tiden mäts från midnatt, och vi vill ha en enklare funktion, kan vi skriva om funktionen som:
$f(t) = 40 + 3t - 36$
detta förenklar till:
$f(t) = 4 + 3t$
Detta är den matematiska formuleringen av djupet vid bryggan som funktion av tiden t.
Hur tycker du att vi gjorde? Vill du försöka att lösa något annat problem?

User: Hej

AI: Hej! Välkommen tillbaka! Hur kan jag hjälpa dig idag? Har du ett nytt matematikproblem som du vill lösa, eller behöver du hjälp med något annat? Jag är här för att hjälpa dig!

User: alice kör 100 m på 10 s enligt funktionen f(x) = x^2 där f(x) är sträckan i meter och x är tiden i sekunder. A) beräkna differenskvoten (f(4,2) - f(4))/(4,2-4)

AI: Låt oss beräkna differenskvoten.
Vi har funktionen $f(x) = x^2$, och vi vill beräkna differenskvoten mellan $x = 4$ och $x = 4,2$.
Först måste vi beräkna $f(4)$ och $f(4,2)$:
$f(4) = 4^2 = 16$ $f(4,2) = (4,2)^2 = 17,64$
Nu kan vi beräkna differenskvoten:
$(f(4,2) - f(4))/(4,2 - 4) = (17,64 - 16)/(4,2 - 4) = 1,64/0,2 = 8,2$
Så, differenskvoten är $8,2$.
Detta är ett sätt att approximera derivatan av funktionen $f(x) = x^2$ vid $x = 4$. Om vi skulle beräkna derivatan av funktionen, skulle vi få:
$f'(x) = 2x$ $f'(4) = 2(4) = 8$
Som du ser, är differenskvoten $8,2$ nästan lika med derivatan 8. Detta beror på att differenskvoten är en approximation av derivatan, och ju närmare x-värdena är, desto bättre är approximationen.
Hur tycker du att vi gjorde? Vill du försöka att lösa något annat problem?

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